In a Different Voice

Gender Composition and Scientific Writing in Economics

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Scientific writing is not a neutral container for ideas

Motivation

- How research is framed shapes peer evaluation and knowledge dissemination
- Gilligan (1993): dominant standards may overlook different modes of communication
- Economics literature focuses on publication and promotion outcomes, not the writing itself

This paper

- Does author-team gender composition predict systematic differences in how economists write?
- Builds on Hengel (2022): female-authored papers are more readable; writing gap widens during peer review
- Extends beyond readability to **15 additional** writing dimensions
- 9,117 abstracts · 16-item LLM rubric

This paper sits at the intersection of two literatures

Gender, voice, & communication

- Gilligan (1993): evaluation standards may embed preferences for particular exposition styles

Gender and economics research

- Card et al. (2020): editors and referees treat submissions differently by gender
- Sarsons (2017): collaborative work differentially credited by gender
- Hengel (2022): female-authored papers more readable; writing gap widens during peer review

Our contribution

- Focus on writing as an observable input into evaluation
- Multidimensional rubric: separates **clarity and structure** from **tone, assertiveness, and affect**
- Three-step design: validation $\rightarrow$ mean differences $\rightarrow$ controlled regressions

9,117 abstracts from four top economics journals, 1950–2015

9,117
abstracts

AER ECA JPE QJE
1950–2015

$<10\%$ female authors overall

$\sim 3\%$ all-female articles

$\sim 9\%$ majority-female

$\sim 13\%$ with ≥ 1 female author

Source: Hengel (2022) replication package (GitHub)

Article data

Title, abstract, citation count, submission/acceptance dates, JEL codes, pre-computed readability scores

Author data

Gender, native language, institution (linked at author level)

Excludes ReSTUD (submission dates only) and AER P&P (no abstracts).

16-item LLM rubric: surface-level linguistic features only

Each criterion scored 1–10. No author intent or personality inferred.

G1 Creativity & Hedging

Modal Verb, Hedging, Qualifier Density, Ack. of Limitations, Caution

G2 Assertiveness & Voice

Assertiveness, Active/Passive Voice

G3 Structural Directness

Sentence Directness, Imperative-Form Occurrence

G4 Authorial Stance & Novelty

Pronoun Commitment, Novelty Claims, Jargon Density, Emotional Valence

G5 Support & Impact

Evidence & Citation Usage, Practical Orientation

Standalone Readability

GPT-5 with chain-of-thought reasoning scored all 9,117 abstracts

- **Model:** GPT-5 — extended reasoning before answering improves accuracy on complex tasks (Shojaee et al. 2025)
- **Pipeline:** abstract $\rightarrow$ rubric $\rightarrow$ model $\rightarrow$ structured 16-score output
- **Prompt engineering:** step-by-step reasoning induced per criterion; shorter inputs increase accuracy (Modarressi et al. 2025)
- **Validation:** trained on subset, tested on holdout, then validated against Hengel (2022) readability scores
- *Caveat:* some randomness in LLM outputs; reported SEs may understate true uncertainty

Three-step analysis: validate → mean differences → controlled regressions

(1) Benchmark LLM readability against Hengel (2022) measures **(2)** Unconditional mean differences across all 16 criteria **(3)** Controlled regressions

$$R_j = \beta_0 + \beta_1 \cdot \mathit{FemaleRatio}_j + \gamma X_j + \delta Y_j + \varepsilon_j$$

X_j: journal×year FE, editor controls, blind-review indicator, JEL codes, theory/empirical flag

Y_j: institution FE, author prominence (max T FE), author seniority (max t FE), native-speaker indicator

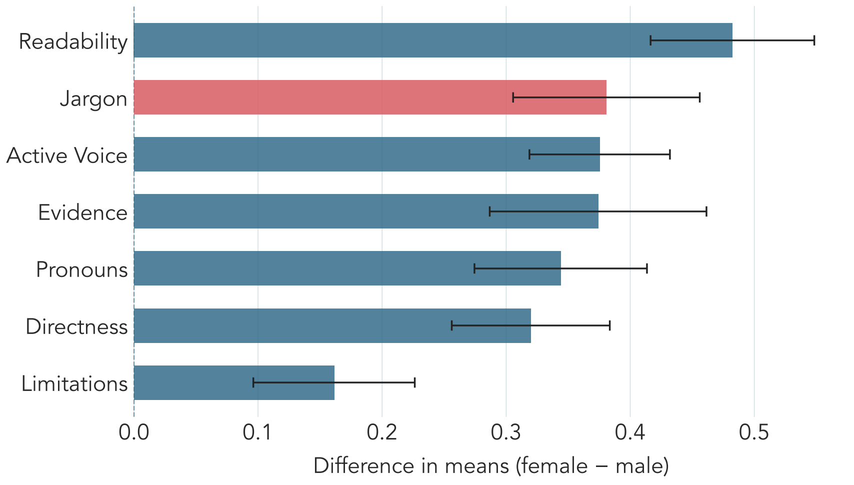
Nine specifications. SEs clustered on editor. Alternative authorship measures: majority-female binary, any-female binary, 100% female, solo, senior female.

LLM readability replicates Hengel's established pattern across all nine specifications

Readability measure	(4)	(5)
Flesch Reading Ease	1.26^{**}	1.49^{***}
Flesch-Kincaid Grade Level	0.24^{*}	0.26^{*}
Gunning Fog Index	0.40^{**}	0.42^{***}
SMOG Index	0.27^{**}	0.29^{**}
Dale-Chall Score	0.08	0.09
LLM Readability	$\mathbf{0.40}^{***}$	$\mathbf{0.32}^{***}$
N	9,117	9,117

OLS coefficients on female ratio. SEs clustered on editor. Col. (4): journal×year, N_j, institution, editor, blind-review. Col. (5) adds quality controls, native speaker, JEL, theory/empirical.

Female-majority teams score higher on clarity; tonal differences are negligible



Controlled regressions confirm the clarity differential; tonal dimensions remain null

Criterion	(4)	(5)
Jargon/Technicality Density[†]	0.55^{***}	0.48^{***}
Sentence Length & Directness	0.31^{***}	0.25^{**}
Evidence & Citation Usage	0.46^{***}	0.36^{***}
Pronoun Commitment	0.39^{**}	0.32
Acknowledgement of Limitations	0.24^{**}	0.15
Active/Passive Voice	0.16	0.15
N	9,117	9,117

OLS coefficients on female ratio. SEs clustered on editor. Col. (4): journal×year, N_j, institution, editor, blind-review. Col. (5) adds quality controls, native speaker. [†]Scale-flipped (11−raw); positive = less jargon.

The clarity differential is robust across six alternative authorship specifications

What we vary

- FemaleRatio (continuous) — main spec
- Majority-female binary ($\geq 50\%$)
- Any-female binary (≥ 1 female author)
- All-female binary (100% female)
- Solo-authored articles only
- Senior female coauthor vs. all-male papers

What holds

- Readability differential significant at 1% in all six specifications
- Jargon, directness, and evidence differentials similarly robust
- Tonal dimensions (assertiveness, hedging, emotional valence) remain null across all specifications
- Results consistent in full sample (1950–2015) and restricted sample (1990–2015)

Female-majority teams in economics write more clearly.

LLM readability validates against Hengel (2022). In mean differences and controlled regressions, female-majority teams score higher on readability, directness, active voice, and evidentiary support, and lower on jargon and novelty claims. Assertiveness, hedging, and emotional valence show no systematic difference.

Backup: Summary statistics by gender composition

	All-Male ($N = 7{,}951$)	Mixed ($N = 853$)	All-Female ($N = 313$)	Full ($N = 9{,}117$)
Readability	6.70 (1.38)	7.42 (1.25)	7.37 (1.30)	6.79 (1.38)
Directness	5.94 (2.08)	6.71 (1.81)	6.53 (1.86)	6.03 (2.06)
Jargon (raw)	7.00 (1.73)	6.48 (1.86)	6.07 (1.92)	6.92 (1.76)
Evidence	1.68 (1.34)	2.31 (1.90)	2.19 (1.73)	1.75 (1.43)
Novelty	2.27 (2.12)	2.26 (2.20)	1.90 (1.85)	2.25 (2.12)
Assertiveness	8.18 (1.55)	8.43 (1.28)	8.04 (1.55)	8.20 (1.53)
Hedging	8.35 (1.98)	8.49 (1.73)	8.07 (2.11)	8.35 (1.96)
Emotional val.	1.12 (0.37)	1.11 (0.35)	1.10 (0.31)	1.12 (0.37)

Means and SDs. Scored 1–10. Descriptive only; no significance stars.

Backup: Key regression coefficients (col. 1, full sample)

Criterion	Coef.	SE
Female > male (significant)		
Jargon (less dense)	$+0.52^{***}$	(0.11)
Evidence & citation	$+0.44^{***}$	(0.09)
Readability	$+0.36^{***}$	(0.07)
Directness	$+0.27^{**}$	(0.10)
Female < male (significant)		
Novelty claims	-0.32^{***}	(0.10)
Not significant		
Assertiveness	-0.08	(0.08)
Emotional valence	-0.02	(0.02)

OLS, female ratio, 1950–2015. SEs clustered on editor. $N = 9{,}117$.

Backup: Five clarity dimensions elevated; tonal dimensions negligible

Female-Majority Higher		Statistically Negligible
Readability	$+0.55$ SD	Assertiveness
Jargon density (lower)	$+0.48$ SD	Hedging frequency
Active/passive voice	$+0.44$ SD	Qualifier density
Sentence directness	$+0.38$ SD	Modal verb strength
Evidentiary support	$+0.35$ SD	Emotional valence
Novelty-claim strength	-0.05 SD	Caution-signaling connectors

Unconditional; majority-gender classification; all criteria standardized.

Backup: Identification concern — GPT-5 may introduce systematic measurement bias

Potential confound

GPT-5 may assign systematically

higher scores to abstracts by female-majority teams, independent of underlying linguistic content.

Response

(1) LLM readability replicates Hengel's

manually calculated scores.

(2) Results are consistent across six authorship specifications.

(3) A model exhibiting such bias would produce elevated scores on assertiveness and hedging. The data show no such pattern.